

SM-Channel Identity Layer

VSEPR-SIM — Theory, Mathematics, and Code Reference

include/IDENTITY/ include/PARTICLES/ include/COLLISION/

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Motivation and Design Philosophy

Classical particle simulations traditionally compute interactions as a single universal force rule evaluated over every pair. A scalar force magnitude is produced, integrated, and repeated. This correctly models billiard-ball mechanics and is inadequate for anything more interesting.

The Standard Model does not work this way. It treats particle interactions as *allowed transitions between field states*, constrained by conservation laws, quantum number compatibility, and the discrete set of force carriers available for a given pair. A photon is exchanged, or a W boson is, or a gluon is. Which carrier mediates the interaction determines everything: whether identity is preserved, mutated, branched, confined, or annihilated.

The SM-channel identity layer introduces this logic into VSEPR-SIM. It does not replace the Cornell QCD force kernel. It wraps around it as a gate layer, selecting which physics fires per pair and recording what identity information is gained or lost in the process.

Principle 1 (Channel Selection). Particle-particle interactions are modeled as gated transitions between recoverable identity-information states. Standard Model-inspired channels determine whether identity is preserved, transformed, branched, confined, or annihilated.

IKK theoretical grounding. The Inverse Kaluza-Klein (IKK) framework (IKK_Doc_I, IKK_Doc_II) formalises the same intuition at a more fundamental level: the observed wavefunction $\Psi(\mathbf{r}, t)$ is a *scale-projected shadow* of a fuller identity structure $\mathcal{I}$ defined over a scale ladder $\{S_0, S_1, \dots, S_D\}$. Failures of the standard equation at high energy, many-body, and cosmological scales are *projection losses* — physics present in $\mathcal{I}$ but not captured by the S_3/S_4 projection. The SM-channel layer is the VSIM operationalisation of this idea: each particle carries a `scale_mask` bitmask naming which scale levels are active, and hidden-intensity fields (H_{charge} , H_{color} , H_{spin}) track structure invisible to the net-zero observable channel. See `section_ikk_kernel_bridge` for the full theoretical-to-code mapping.

Notation and Definitions

Particles, Channels, and Output

Let C_i denote the *complete state* of particle i : its position, velocity, and full identity bundle. Let I_i denote the *recoverable identity-information* scalar for particle i . An interaction between particles i and j mediated by channel A_{ij} is:

$$C_i + C_j \xrightarrow{A_{ij}} I_{\text{out}} \quad (\text{scattering / identity-preserving}) \quad (1)$$

$$I_i + I_j \xrightarrow{A_{ij}} C_{\text{products}} \quad (\text{annihilation}) \quad (2)$$

Interaction Channel Row

For a pair (i, j) , the full channel row M_{ij} is:

$$M_{ij} = [m_\gamma, m_g, m_W, m_Z, m_H, m_{\text{ann}}, m_{\text{pair}}, m_{\text{decay}}, m_{\text{conf}}]_{ij} \quad (3)$$

The aggregate interaction amplitude is:

$$A_{ij} = \sum_k M_{ij}^{(k)} K_k(C_i, C_j, I_i, I_j, Z_i, Z_j) \quad (4)$$

Table 1 defines each channel.

Table 1: SM-inspired interaction channels in M_{ij} .

Symbol	SM analogy	Status	What is logged
m_γ	EM / photon-mediated	active	charge, scattering, radiation
m_g	Strong / gluon-mediated	active	color flow, Cornell gate
m_W	Weak charged current	gated/zero	flavor/charge-changing transition
m_Z	Weak neutral current	gated/zero	weak scattering, no charge exchange
m_H	Higgs / scalar coupling	gated/zero	mass-coupling interaction
m_{ann}	Annihilation	active	particle-antiparticle collapse score
m_{pair}	Pair production	gated/zero	energy-to-particle creation
m_{decay}	Unstable decay	gated/zero	lifetime, branching ratio
m_{conf}	Confinement/hadronization	gated/zero	no free color states

The Z -Matrix

Each particle carries two identity matrices:

Definition 1 (Z -matrix). $Z_i^{(2)} \in \mathbb{R}^{2 \times 2}$ is the *runtime state matrix* of particle i . It encodes the particle's current field configuration and is updated by the integrator after each interaction step.

$Z_i^{(3)} \in \mathbb{R}^{3 \times 3}$ is the *fundamental identity matrix* in the three-generation space. It changes only during weak-mediated identity mutation events.

Both matrices are plain-aggregate types storing `double[n][n]` in row-major order. The key operation is *matrix opposition*:

$$\kappa_Z(Z_i, Z_j) = 1 - \frac{\langle Z_i, Z_j \rangle_F}{\|Z_i\|_F \|Z_j\|_F + \varepsilon} \quad (5)$$

where $\langle A, B \rangle_F = \sum_{r,c} A_{rc} B_{rc}$ is the Frobenius inner product. The range is $[0, 1]$: zero means the matrices are identical (same state), one means they are fully opposed (antiparticle pairing).

An alternative form via the trace product is:

$$\kappa_Z = 1 - \frac{\text{tr}(Z_i Z_j)}{\|Z_i\|_F \|Z_j\|_F + \varepsilon} \quad (6)$$

IKK connection: hidden-active condition. IKK Doc I (§3.3) and Doc II (§D) establish that a net-zero observable channel ($I_j = \alpha^\top \mathbf{c}_j = 0$) absence of internal structure. The hidden intensity $H_j = \mathbf{c}_j^\top \mathbf{W} \mathbf{c}_j > 0$ can coexist with $I_j = 0$. Canonical example: a neutral baryon (up+down+down quarks) has $I_{\text{charge}} = +\frac{2}{3} - \frac{1}{3} - \frac{1}{3} = 0$ but $H_{\text{charge}} = \frac{2}{3} \neq 0$. The `ParticleIdentity` fields `H_charge`, `H_color`, `H_spin` and the `hidden_active_charge()` member implement this detection. The $Z3$ 3×3 matrix is the direct analog of the quark component matrix $\mathbf{C}_q$ from IKK Doc II (§C).

Conservation Gates

Every interaction is conditional on a conservation ledger. The gate product is:

$$\mathcal{G}_{\text{cons}} = G_E \cdot G_p \cdot G_q \cdot G_\ell \cdot G_b \cdot G_c \cdot G_s \quad (7)$$

Table 2 defines each gate.

Table 2: Conservation gates and their checks.

Gate	Quantity	Condition checked
G_E	Energy conservation	$E_{\text{rel}} \geq 0$, kinematically allowed
G_p	Momentum balance	$\Delta p/p_{\text{total}} < \epsilon_p$
G_q	Electric charge	$\sum q$ is finite and consistent
G_ℓ	Lepton family	families match or at least one is zero
G_b	Baryon number	additive, tracked per interaction
G_c	Color neutrality	colored pairs satisfy $c_j = \bar{c}_i$
G_s	Spin compatibility	spin types are pairwise compatible

The annihilation condition requires all gates plus spatial/kinematic gates:

$$A_{ij} = G_{\text{cons}} \cdot G_{\text{kin}} \cdot G_{\text{overlap}} \cdot G_\kappa \cdot G_L \quad (8)$$

where:

$$G_{\text{overlap}} : r_{ij} < r_c \quad (9)$$

$$G_{\text{kin}} : E_{\text{rel}} > E_c \quad (10)$$

$$G_\kappa : \kappa_{ij}^{\text{total}} > \kappa_c \quad (11)$$

$$G_L : L_{ij} > L_{\text{ann}} \quad (12)$$

Kappa: Compatibility and Opposition Scores

Sub-scores

Five orthogonal compatibility scores are computed per pair:

$$\kappa_Z = 1 - \frac{\langle Z_i, Z_j \rangle_F}{\|Z_i\|_F \|Z_j\|_F + \varepsilon} \quad (\text{matrix opposition})$$

$$\kappa_q = 1 - \frac{|q_i + q_j|}{|q_i| + |q_j| + \varepsilon} \quad (\text{charge complementarity})$$

$$\kappa_s = \begin{cases} 1 & s_i = s_j \\ 0.5 & \text{otherwise} \end{cases} \quad (\text{spin compatibility})$$

$$\kappa_c = \begin{cases} 1 & c_j = \overline{c_i}, \text{ both colored} \\ 0 & \text{otherwise} \end{cases} \quad (\text{color/anticolor match})$$

$$\kappa_E = 1 \quad (\text{energy availability (placeholder)})$$

Weighted Total

$$\boxed{\kappa_{ij}^{\text{total}} = w_Z \kappa_Z + w_q \kappa_q + w_s \kappa_s + w_c \kappa_c + w_E \kappa_E} \quad (13)$$

Default weights in `ChannelPolicy`:

Weight	w_Z	w_q	w_s	w_c	w_E
Value	0.40	0.30	0.10	0.10	0.10

Identity Information and State Loss

Recoverable Identity Information

Each particle carries a scalar $I_i \in [0, 1]$ representing how much of its original identity information remains recoverable. A freshly created particle has $I_i = 1$. After interactions:

Definition 2 (State Loss Principle).

$$\Delta I_{ij} = I_i^+ - I_i^- \quad (\text{change over one interaction}) \quad (14)$$

$$L_{ij} = -\Delta I_{ij} \quad (\text{state loss, } \geq 0 \text{ for ordinary loss}) \quad (15)$$

The three regimes:

Condition	Name	Meaning
$\Delta I_{ij} < 0$	Ordinary loss	Identity information is dissipated
$\Delta I_{ij} = 0$	Identity-preserving	Elastic or neutral-current scattering
$\Delta I_{ij} > 0$	Positive event	Short-lived structured state forming

The Annihilation Threshold

$$G_L = \begin{cases} 1 & L_{ij} > L_{\text{ann}} \\ 0 & L_{ij} \leq L_{\text{ann}} \end{cases} \quad \text{where } L_{ij} = -\Delta I_{ij} \quad (16)$$

Positive Information Events

When $\Delta I_{ij} > 0$, a *positive information event* is flagged. This means:

$$\boxed{\text{Recoverable identity temporarily becomes more structured before dissipating.}} \quad (17)$$

Physical interpretations:

Event	Meaning
Temporary pairing	Short-lived structured two-particle state
Resonance	Intermediate state before decay
Matrix alignment	(Z_i, Z_j) become unusually compatible
Pre-annihilation order	Final collapse is likely
Confinement ordering	Colored components forming neutral products
Shower coherence	Branching with organized angular/color structure

ΔI Heuristic Kernel

The current ΔI computation is a heuristic pending a full branching-ratio kernel. It decomposes into four contributions:

$$\begin{aligned}
\Delta I^{\text{EM}} &= -0.05 m_\gamma && \text{(small EM loss)} \\
\Delta I^{\text{ann}} &= -0.90 m_{\text{ann}} && \text{(near-total annihilation loss)} \\
\Delta I^{\text{color}} &= -0.20 m_g \kappa_c && \text{(color exchange loss)} \\
\Delta I^{\text{res}} &= +0.10 (1 - \kappa_Z) \cdot \mathbf{1}[(1 - \kappa_Z) > 0.80] && \text{(resonance / alignment gain)}
\end{aligned}$$

Total, clamped to $[-I_i, +0.15 I_i]$:

$$\Delta I_{ij} = \text{clamp}\left(\Delta I^{\text{EM}} + \Delta I^{\text{ann}} + \Delta I^{\text{color}} + \Delta I^{\text{res}}, -I_i, 0.15 I_i\right) \quad (18)$$

Full Annihilation Gate Stack

Electron-Positron Prototype

The canonical annihilation prototype in VSEPR-SIM:

$$e^- + e^+ \xrightarrow{A} \gamma_1 + \gamma_2 \iff I_{e^-} + I_{e^+} \xrightarrow{A} C_{\gamma_1} + C_{\gamma_2} \quad (19)$$

Required conditions:

$$q_i + q_j = 0 \quad p_i + p_j = p_{\text{out}} \quad E_i + E_j = E_{\text{out}} \quad \kappa_{ij}^{\text{total}} > \kappa_c \quad (20)$$

Gate Stack Summary

$$\text{annihilation_candidate} = \underbrace{r < r_c}_{G_{\text{overlap}}} \wedge \underbrace{E_{\text{rel}} > E_c}_{G_{\text{kin}}} \wedge \underbrace{\kappa^{\text{total}} > \kappa_c}_{G_{\kappa}} \wedge \underbrace{L_{ij} > L_{\text{ann}}}_{G_L} \wedge \underbrace{\mathcal{G}_{\text{cons}}}_{G_{\text{cons}}} \quad (21)$$

$$\text{annihilation_confirmed} = \text{candidate} \wedge |q_i + q_j| < 10^{-6} \quad (22)$$

Default threshold values in `ChannelPolicy`:

Parameter	Field	Default	Meaning
r_c	<code>r_c</code>	0.20	max separation ($\text{\AA}$ analog)
E_c	<code>E_c</code>	0.50	min relative energy
κ_c	<code>kappa_c</code>	0.70	min κ^{total}
L_{ann}	<code>L_ann</code>	0.50	min state loss

Strong Interaction and the Color Channel Gate

The Cornell QCD kernel fires only when the color channel m_g exceeds the activation threshold δ_g :

$$\text{fire_cornell} = \begin{cases} 1 & m_g > \delta_g \text{ (both particles color-charged)} \\ 0 & \text{otherwise} \end{cases} \quad \delta_g = 0.05 \quad (23)$$

Scale-Dependent Treatment

Scale	Treatment
Very near	Direct color/matrix interaction via κ_c
Intermediate	Cornell string/flux-tube proxy
Far	Color-neutral aggregate only
Post-collision	Shower + hadronization event model

Octree Color Neutrality Requirement

QCD octree aggregation must track color neutrality, not just charge and mass.

 (24)

The planned `OctreeIdentityNode` adds to the existing Barnes–Hut tree:

- `color_neutrality_score` — fraction of color-neutral aggregate
- `confinement_risk` — probability of forced confinement in one T_B step
- `shower_probability` — probability of parton-shower initiation

Weak Interaction and Identity Mutation

Weak charged-current interactions can change particle type. This is the clean justification for an event where:

$$I_i^- \neq I_i^+ \quad (\text{identity mutation, not just identity loss}) \quad (25)$$

The beta-decay prototype in VSEPR-SIM notation:

$$C_d \xrightarrow{A_W} C_u + C_{W-} \quad C_{W-} \rightarrow C_{e-} + C_{\bar{\nu}_e} \quad (26)$$

The `weak_identity_transition` flag in `InteractionCandidate` is set when $m_W > 0$. The W kernel is gated at zero in v5.2.0; the flag exists for forward-compatibility with the live event logger.

Code Reference: Data Structures

Matrix Types (IDENTITY/identity_matrix.hpp)

```

1 struct Matrix2x2 {
2     double m[2][2]{};
3     double& operator()(int r, int c);
4     static Matrix2x2 zero();
5 };
6
7 struct Matrix3x3 {
8     double m[3][3]{};
9     double& operator()(int r, int c);
10    static Matrix3x3 zero();
11 };
12
13 // Free functions (same family for Matrix3x3):
14 Matrix2x2 identity_2x2();
15 double trace(const Matrix2x2&);
16 double frobenius_norm(const Matrix2x2&);
17 double frobenius_dot(const Matrix2x2&, const Matrix2x2&);
18 double opposition(const Matrix2x2&, const Matrix2x2&,
19                  double eps = 1e-15);
20 double matrix_compatibility(const Matrix2x2&, const Matrix2x2&,
21                             double eps = 1e-15);
22 Matrix2x2 mat_mul(const Matrix2x2&, const Matrix2x2&);

```

Listing 1: Matrix2x2 and Matrix3x3 plain aggregates.

ParticleIdentity (IDENTITY/particle_identity.hpp)

```

1 struct ParticleIdentity {
2     int64_t id {0};
3     int32_t type_code {0};
4
5     double charge {0.0};
6     double mass {0.0};
7     SpinProxy spin_proxy {SpinProxy::Half};
8
9     particles::ColorCharge color {ColorCharge::Neutral};
10    particles::ColorCharge anti_color {ColorCharge::Neutral};
11
12    int8_t lepton_family {0}; // 0=none, 1=e, 2=mu, 3=tau
13    int8_t baryon_number {0};
14    int8_t isospin_proxy {0};
15
16    Matrix2x2 Z2; // runtime 2x2 state matrix
17    Matrix3x3 Z3; // 3x3 generation/flavor identity matrix

```



```

18     double    I_recoverable    {1.0};
19     uint64_t  identity_hash    {0};
20
21
22     void refresh_hash() noexcept;    // FNV-1a 64-bit over quantum-
        number fields
23 };
24
25 // Convenience constructors:
26 ParticleIdentity make_electron_like (int64_t id, double mass =
        0.000511);
27 ParticleIdentity make_positron_like (int64_t id, double mass =
        0.000511);
28 ParticleIdentity make_quark_like     (int64_t id, QuarkFlavor,
        ColorCharge);

```

Listing 2: ParticleIdentity struct.

InteractionChannelRow (IDENTITY/interaction_channel.hpp)

```

1 struct InteractionChannelRow {
2     double m_gamma {0.0};    // EM (active)
3     double m_g      {0.0};    // strong (Cornell gate)
4     double m_W      {0.0};    // weak charged (gated/zero)
5     double m_Z      {0.0};    // weak neutral (gated/zero)
6     double m_H      {0.0};    // Higgs (gated/zero)
7     double m_ann     {0.0};    // annihilation (active)
8     double m_pair    {0.0};    // pair prod. (gated/zero)
9     double m_decay   {0.0};    // decay (gated/zero)
10    double m_conf     {0.0};    // confinement (gated/zero)
11
12    bool color_channel_active(double threshold = 0.05) const;
13 };
14
15 struct KappaScores {
16     double kappa_Z, kappa_q, kappa_s, kappa_c, kappa_E;
17 };
18
19 struct ConservationGates {
20     bool G_E, G_p, G_q, G_l, G_b, G_c, G_s;
21     bool conservation_pass() const;    // AND of all gates
22 };
23
24 struct ChannelPolicy {
25     double w_Z{0.40}, w_q{0.30}, w_s{0.10}, w_c{0.10}, w_E{0.10};
26     double r_c{0.20}, E_c{0.50}, kappa_c{0.70}, L_ann{0.50};
27     double m_g_threshold{0.05};
28 };

```

Listing 3: Nine-channel row and associated types.

InteractionCandidate (IDENTITY/interaction_candidate.hpp)

```

1 struct InteractionCandidate {
2     int64_t i, j;    // particle indices
3     double r;    // separation
4     double E_rel;    // relative kinetic energy

```

```

5     double    p_balance_error;    // |delta_p| / |p_total|
6
7     KappaScores      kappa;
8     double           kappa_total_val;
9     InteractionChannelRow channels;
10    ConservationGates gates;
11    bool              conservation_pass;
12
13    double    delta_I;           // I_out - I_in
14    double    state_loss;        // L_ij = -delta_I  (>= 0)
15
16    double    annihilation_score;
17    double    shower_score;
18    double    weak_transition_score;
19    double    resonance_score;
20
21    bool      annihilation_candidate;
22    bool      annihilation_confirmed;
23    bool      positive_information_event;
24    bool      weak_identity_transition;
25    bool      color_exchange_active;
26    bool      temporary_resonance;
27 };
28
29 // Main evaluator -- stateless, allocation-free.
30 InteractionCandidate evaluate_pair(
31     const ParticleIdentity& a,
32     const ParticleIdentity& b,
33     const ChannelPolicy&    policy,
34     double r, double E_rel,
35     double p_balance_error = 0.0);

```

Listing 4: Full pair evaluation record and evaluator.

Cornell Color-Channel Gate (COLLISION/qcd_force.hpp)

```

1 // SM-gated overload. Cornell fires only when m_g > policy.
  // m_g_threshold.
2 // Original qcd_accumulate_forces() is untouched.
3 void qcd_accumulate_forces_sm(
4     particles::QuarkGluonPlasma&          plasma,
5     std::vector<identity::ParticleIdentity>& identity_layer,
6     const identity::ChannelPolicy&        policy,
7     std::vector<identity::InteractionCandidate>*& candidates_out);

```

Listing 5: qcd_accumulate_forces_sm() – SM-gated force pass.

Planned Event Types for Live Logger

The live event logger (next pipeline stage) maps directly onto the `InteractionCandidate` flag fields.

Event type string	Triggering condition
elastic_scattering	$\Delta I \approx 0$, no channel triggered
inelastic_scattering	$\Delta I < 0$, EM active
radiative_emission	$m_\gamma > 0$, significant loss
pair_production	$m_{\text{pair}} > 0$ (future)
annihilation_candidate	annihilation_candidate == true
annihilation_confirmed	annihilation_confirmed == true
weak_identity_transition	$m_W > 0$ (future)
neutral_current_scattering	$m_Z > 0$ (future)
color_exchange	color_exchange_active == true
parton_branch	shower initiated (future)
hadronization_candidate	$r > r_{\text{conf}}$ (Cornell gate)
temporary_resonance	temporary_resonance == true
positive_information_event	$\Delta I > 0$
identity_loss_spike	$L_{ij} \gg L_{\text{ann}}$

Example .ee Event Records

```

frame 2091
event annihilation_candidate
particles 4001 4002
channel e_minus_e_plus_to_gamma_gamma
kappa_total 0.941
delta_I -0.882
state_loss 0.882
conservation_pass true
annihilation_score 0.918
end

```

```

frame 884
event temporary_resonance
particles 102 103
kappa_Z 0.031
delta_I +0.087
state_loss 0.000
positive_information_event true
end

```

Summary

Particle-particle interactions are modeled as gated transitions between recoverable identity-information states, where Standard Model-inspired channels determine whether identity is preserved, transformed, branched, confined, or annihilated.

The three state-loss identities:

$$\begin{aligned}
\Delta I_{ij} \leq 0 &\Rightarrow \text{ordinary interaction loss} & (27) \\
L_{ij} = -\Delta I_{ij} &\Rightarrow \text{measured state loss} & (28) \\
\Delta I_{ij} > 0 &\Rightarrow \text{short-lived positive information event} & (29)
\end{aligned}$$

File and Library Reference

File	Contents
IDENTITY/identity_matrix.hpp	Matrix2x2, Matrix3x3, free helpers
IDENTITY/particle_identity.hpp	ParticleIdentity, SpinProxy, constructors
IDENTITY/interaction_channel.hpp	Channel row, gates, kappa scoring
IDENTITY/interaction_candidate.hpp	InteractionCandidate, evaluate_pair()
IDENTITY/identity.hpp	Umbrella include
PARTICLES/particles.hpp	Extended: includes IDENTITY/identity.hpp
COLLISION/qcd_force.hpp	Extended: qcd_accumulate_forces_sm()

Dependency chain (header-only, single direction):

identity_matrix $\longrightarrow$ particle_identity $\longrightarrow$ interaction_channel $\longrightarrow$
interaction_candidate

IKK theory documents (companion reading):

- **IKK_Doc_I_Schrodinger_Alternative** — scale-ladder definition, identity bundle, hidden-active condition, projection failure, entanglement relation-particle.
- **IKK_Doc_II_Initial_Draft** — scale axioms, matrix packing chain, entropy and complexity, dark matter kernel equations, Python and C++ prototypes, open variable register.
- **IKK_Doc_III_Kernel_V4_Projection** — bilinear projection form for radial density.
- **section_ikk_kernel_bridge** — formal mapping of IKK theory to VSIM IDENTITY/ kernel: ScaleLevel, scale_mask, fuzz_ratio(), H.charge/color/spin, hidden-active flags, open variable register.